

The empirical recurrence for OEIS A296685: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A296685 records “Empirical recurrence of order 63”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 115$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A296685 is “Number of nX5 0..1 arrays with each 1 adjacent to 1, 2 or 3 king-move neighboring 1s.”. It has offset 1 and begins

12, 524, 6959, 121477, 2323514, 40828110, 732185986, 13222385649, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Dec 18 2017 12:08:57, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1056$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 115$ of the 1056, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 115$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 230$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	11	c_{17}	−13041597081	c_{33}	1573262494537	c_{49}	−641712953518
c_2	46	c_{18}	−7396897862	c_{34}	−9349367847173	c_{50}	357726593822
c_3	1338	c_{19}	40423671336	c_{35}	−12362107500000	c_{51}	444593163354
c_4	623	c_{20}	145005733027	c_{36}	−3716911798736	c_{52}	63520025166
c_5	22346	c_{21}	85294806902	c_{37}	6409805004386	c_{53}	−64397834316
c_6	−84638	c_{22}	−368492248905	c_{38}	11886766910674	c_{54}	−21761666540
c_7	−112316	c_{23}	−577230820031	c_{39}	8294460079252	c_{55}	4237347662
c_8	−2083625	c_{24}	−108660182013	c_{40}	−4249996122171	c_{56}	3285830008
c_9	−3421046	c_{25}	770811635618	c_{41}	−12880350222825	c_{57}	−301206994
c_{10}	−9732131	c_{26}	2154568905525	c_{42}	−8761686352904	c_{58}	−214074312
c_{11}	14476964	c_{27}	1150445995883	c_{43}	2831764499911	c_{59}	7184376
c_{12}	105035305	c_{28}	−3842409326692	c_{44}	8516023267537	c_{60}	12969096
c_{13}	379113141	c_{29}	−5699883463474	c_{45}	5835200756377	c_{61}	−1964192
c_{14}	537901601	c_{30}	−184466037292	c_{46}	647674610371	c_{62}	24864
c_{15}	−308515105	c_{31}	7472671763811	c_{47}	−3058538774991	c_{63}	7040
c_{16}	−5458428494	c_{32}	9288544867073	c_{48}	−2423623813721		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63 + 4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A296685.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.